GammaLoop

Differential cross-sections with Local Unitarity

Version source: `pyproject.toml and crates/gammaloop-rs/Cargo.toml`   Owner: GammaLoop project

1 Overview

GammaLoop computes differential collider cross-sections with the Local Unitarity method. It is an application and research codebase rather than a thin numerical library: process generation, model import, integrand construction, integration, observables, persistence, and diagnostics meet in one stateful workflow.

> **Primary interface**
> The command-line interface is the primary user interface. A run card creates a state, executes commands, and leaves a reusable working directory. The Rust and Python APIs expose the same loading boundary and selected structured operations; they do not define a second, independent execution model.

1.1 A stateful run

A typical source checkout is built and exercised from the repository root:

```
just build-cli-release
./gammaloop ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml
```

The run card imports a model, generates the requested process, executes its command blocks, and records the resulting state. The state directory is the durable boundary for subsequent work. Resume it explicitly rather than reconstructing hidden in-memory context:

```
./gammaloop -s ./examples/cli/gg_hhh/1L/state \
  run integrate_physical -c "quit -o"
```

The persisted `run.toml` is an audit and replay description. The settings files describe global and default runtime configuration, while `processes/` contains process-specific state. Ordinary runs create `gammaloop_state/` unless a different state path is supplied.

1.2 Installation and external tools

GammaLoop is currently developed primarily from source. The supported development path is the repository's Nix shell, or a local Rust toolchain together with `just`, a recent GNU toolchain, and Python 3.11 or newer when building bindings. FORM 4.2.1 or newer is needed for analytical integration of integrated UV counterterms. UFO model import also needs the Python `ufo-model-loader` package.

```
nix develop
just build-cli
just build-api
```

Diagram rendering is a separate concern and uses Clinnet and Typst. Building the CLI does not imply that these drawing tools are installed.

1.3 Ownership across the stack

GammaLoop owns workflows, not every dependency concept
Linnet owns the half-edge graph model and graph algorithms. Spenso owns typed tensors, tensor structures, and network execution. Idenso owns symbolic tensor identities and algebraic simplification. Vakint owns vacuum-integral topology matching and evaluation. GammaLoop documents how these components participate in a collider calculation and links to their API references instead of duplicating them.

The maintained implementation overview is the current architecture note. Proposal and status documents are useful development records, but they are not promises about the supported user interface.

1.4 Where to begin

- Use built-in `--help` for the exact CLI surface in the current build.
- Start with the maintained `gg_hhh` run card for a complete state lifecycle.
- Use the Rust or Python API only when a program needs structured access to loaded state or sample-evaluation results.
- Keep expensive integrations and licensed/external tools out of ordinary documentation rendering; examples should declare their prerequisites.

2 Tutorial

This tutorial builds the GammaLoop command-line interface and uses the maintained one-loop `g g -> h h h` run card to create a state that can be inspected and resumed. It is a real process-generation workflow, so expect compilation to take longer than a `--help` smoke test.

2.1 Prerequisites

Work from a GammaLoop source checkout. The supported development environment is the repository Nix shell; without Nix you need Rust 1.85 or newer, `just`, a recent GNU toolchain, and the dependencies described in the root README. The bundled Standard Model used below does not need the UFO loader. A Symbolica license may be required by your build and use case.

```
nix develop
just build-cli
./gammaloop --help
```

The last command should print the one-shot CLI options and command tree. The `./gammaloop` wrapper selects the binary built under `target/dev-optim/`.

2.2 Generate one maintained process

The run card declares a state folder, settings, and named command blocks. Run its `generate` block and persist the result on exit:

```
./gammaloop --clean-state \
  ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml \
  run generate -c "quit -o"
```

`--clean-state` removes the resolved state first
Use it only for this first, reproducible run. Omit it when resuming work you want to keep. The maintained card resolves its state to `examples/cli/gg_hhh/1L/state` and requests ten worker cores; make a private copy of the card before changing either value.

The block imports `sm-default.json`, generates the selected one-loop pentagon contribution, builds its integrand, saves DOT data, and writes the state. A successful run exits without an error and leaves at least `run.toml`, `global_settings.toml`, `default_runtime_settings.toml`, and `processes/` in the state folder. `run.toml` is the audit and replay description; it is not merely a log file.

2.3 Prove that the state is reusable

Load the saved state and ask the active session to display its processes:

```
./gammaloop -s ./examples/cli/gg_hhh/1L/state \
  run -c "display processes; quit -o"
```

Success means the display includes the generated `gg_hhh` process and the command exits while keeping the same state. You can now run the card's `integrate_euclidean` or `integrate_physical` block, but those are deliberately beyond the first-success path because they request a substantial Monte Carlo integration.

2.4 Troubleshooting and next steps

- If the wrapper cannot find a binary, rerun `just build-cli` and confirm that `target/dev-optim/gammaloop` exists.
- If compilation or process generation exhausts local resources, copy the card and reduce the values under `cli_settings.global.n_cores`; do not edit a state midway through generation.
- If startup reports a state fingerprint or settings mismatch, replay the card with a new state folder instead of transplanting only its `processes/` directory.
- Use `./gammaloop help generate` for flags from the exact build you are running, then continue with the process-generation manual and the generated CLI reference.

Ground truth for this walkthrough
The complete card is the maintained gg_hhh example. The executable's parser and generated CLI catalog own option spelling; this tutorial owns the sequence and the meaning of the resulting state.

3 Process generation and state workflows

This chapter replaces the former GammaLoop wiki process page. The old page used the retired `bin/compile.sh`, `bin/gammaloop`, and `.gL` card layout. Current releases use the repository wrapper or Cargo-built executable, TOML run cards, and a persistent state folder. The parser, its tests, and the generated Clap catalog are the authority for spelling in a particular snapshot.

3.1 Choose a generation mode

The interactive command tree separates cross sections, amplitudes, and operations on processes already stored in the active state:

```
generate xs INITIAL > FINAL ...
generate amp INITIAL > FINAL ...
generate existing ...
generate help xs
```

`xs` constructs forward-scattering graphs and filters compatible Cutkosky cuts. `amp` constructs ordinary amplitudes. Their final-state lists therefore have different meanings even when the printed particle syntax is identical. Use `generate help <mode>` inside the stateful CLI for the exact flags available in the selected build.

3.2 Process specification grammar

The source-backed grammar accepts `to` or `>` between space-separated initial and final states. Braces express alternatives in a cross-section final state, while `{}` denotes an empty state:

```
e+ e- > mu+ mu-
e+ e- > { Z Z, a a, H H }
{} > {}
```

A slash vetoes particles and a vertical bar selects an allow-list. The words `veto` and `only` are equivalent forms. Amplitude coupling constraints use the coupling name directly; powered constraints apply to the complete cross section:

```
e+ e- > d d~ g / u c QED==2 QCD>=2 QCD<=4
e+ e- > mu+ mu- | g ghG QED^2==2 QCD^2>=2
```

The bracketed perturbative block is intentionally distinct from ordinary flags. `{n}` fixes the amplitude loop count (or the sum across the two cut sides), `{{n}}` fixes the forward-graph loop count, and `QCD=n` or the shorthand `QCD` selects a relative perturbative order:

```
e+ e- > Z [ {1} {{2}} QCD=2 QED=1 ]
e+ e- > Z [ QCD ]
```

Resolve names through the active model
Particle names, antiparticles, coupling-order names, and allowed interactions come from the model loaded in the current state. A syntactically valid specification can still be rejected when that model does not define a requested name.

3.3 Filters, grouping, and multiplicity

Generation flags control topology filters, loop ranges, cut blobs and spectators, symmetrization, and numerator-aware grouping. These choices are physics-significant. In particular, grouping can combine graph multiplicities and numerator ratios; external-fermion symmetrization requires the caller to retain the associated ordering signs. Do not copy an option list from an older snapshot: use the generated CLI reference, which records aliases, defaults, possible values, and arity directly from Clap.

The generated graph's overall factor keeps separate contributions for automorphisms, internal fermion loops, external-fermion ordering, numerator-independent symmetry grouping, and numerator-dependent grouping. Persist the state and inspect that factor when auditing diagram normalization instead of replacing it with an assumed factorial.

3.4 From generation to evaluation

A reproducible run keeps its stages explicit:

- import or select the model before resolving a process specification;
- generate the process and inspect the retained diagrams;
- generate the requested integrand representation;
- set kinematics, sampling, stability, and subtraction settings;
- evaluate samples or run an integration command block;
- exit with persistence enabled when the state should be resumed.

The state is the audit boundary
`run.toml` records boot settings, reusable command blocks, and top-level commands. Generated process and

integrand artifacts live beside it. Resume with `-s STATE_FOLDER`; replay the run card when a clean reconstruction is required. Avoid moving only one generated subdirectory into a different state, because settings and fingerprints are checked together.

The parser and its executable examples are maintained in the generate command source.

4 Rust, Python, and CLI reference boundaries

Generated-reference contract
Signatures, fields, variants, defaults, feature gates, and source locations come from ordered generated descriptors. Hand-written prose explains intent and workflow; it must not duplicate a signature as a second source of truth. Rust documentation explicitly marked as Typst markup may be evaluated as markup. Python docstrings and all other external metadata are escaped as plain text.

Rust scope: `gammaloop`
Python scope: `gammaloop`

4.1 Rust ownership

The Rust surface is split across two primary packages.

- `gammalooprs` contains the physics implementation, data model, integrands, integration, observables, and lower-level algorithms.
- `gammaloop-api` owns the supported state-loading boundary, session/command integration, CLI assembly, and the PyO3 extension used by the Python distribution.

The stable conceptual entry point is `StateLoadOption::load()`. Load-time options select a state directory, boot card, logging overrides, read-only behavior, and optional settings overrides. Once loaded, callers can create a CLI-style session or use dedicated structured operations. A raw command string is the compatibility fallback, not a replacement for a typed operation where one exists.

```rust
use std::path::PathBuf;
use gammaloop_api::{state::CommandHistory, StateLoadOption};

let mut loaded = StateLoadOption {
    state_folder: Some(PathBuf::from("./state")),
    ..StateLoadOption::default()
}.load()?;

let command = CommandHistory::from_raw_string("display settings global")?;
loaded.cli_session().execute_command(command)?;
```

Generated Rust reference pages must distinguish public supported catalog entries from implementation-public types. Conventional Rustdoc remains available for the complete compiled surface and trait implementations.

4.2 Python packaging

A standalone GammaLoop distribution
The repository root `pyproject.toml` builds the Python distribution named `gammaloop` with Maturin. Users import the public package as `gammaloop`; its compiled implementation module is `gammaloop._gammaloop`. This is separate from the `symbolica.community.*` modules used by Spenso, Idenso, and Vakint. Building or installing GammaLoop does not install those community modules as independent Python packages.

The main Python entry point is `GammaLoopAPI`:

```python
from gammaloop import GammaLoopAPI

gl = GammaLoopAPI(
    state_folder="./state",
    boot_commands_path="./run.toml",
)
gl.run("display settings global")
```

The Python package requires Python 3.11 or newer. `just build-api` is the local development build. Distribution builds may enable the stable ABI feature; that choice changes binary compatibility, not the public module name.

4.3 Sample evaluation and precision

`evaluate_sample` returns one sample result and the observable bundle for that one-sample batch. `evaluate_samples` returns per-sample results plus a batch-global observable bundle. Both the ordinary Rust and Python endpoints expose numeric results through an `f64` contract, even if arbitrary precision was used internally. Rust-only callers that must retain the active precision use `evaluate_sample_precise` and `evaluate_samples_precise`.

The generated reference must present these as distinct contracts. It must not suggest that the Python API returns arbitrary-precision numeric values.

4.4 CLI and settings metadata

The executable command tree is generated from Clap and settings descriptions are generated from the existing schema/default machinery. These generated catalogs own command names, aliases, flags, positionals, defaults, possible values, and settings paths. Hand-written guides own sequencing, state effects, and examples.

Source starting points are gammaloop-api and gammalooprs.

5 Releases and change history

> **Release-history policy**
> GammaLoop does not currently have a canonical root `CHANGELOG.md`. Product releases are identified by the root Python version, the `gammalooprs` package version, and repository tags. This manual must not synthesize detailed historical entries from commit messages or reuse the changelogs of Linnet, Spenso, Idenso, or Vakint.

The documentation build may publish immutable snapshots for existing GammaLoop tags and a moving `latest` channel. A snapshot records the repository revision and all component versions from the registry. If release notes are curated later, they should describe user-visible CLI, state, Rust API, Python API, and physics-result changes, including migration requirements.

Until a canonical changelog is added, consult the repository's tags and pull requests and verify behavior against the current source. The short README remains the installation and entry-point summary; architectural truth belongs in architecture-current.md.

Supported API catalog

This curated catalog is generated from explicit ordered scopes. Links lead to the exhaustive language references.

`gammalooprs`

GammaLoop's supported Rust entry points. The complete compiled surface remains available in the Rustdoc sidecar.

`GammaLoopContext`

`Combines the state-map and model capabilities required by GammaLoop computations.`

`Combines the state-map and model capabilities required by GammaLoop computations.`

```
pub trait GammaLoopContext : HasStateMap + HasModel {}
```

Supported usage

```rust
fn accepts_context<C: gammalooprs::GammaLoopContext>(_context: &C) {}
```

Exhaustive reference · Source

`HasModel`

`Provides shared access to the physics model owned by a computation context.`

`Provides shared access to the physics model owned by a computation context.`

```
pub trait HasModel { fn get_model(& self) -> & Model; }
```

Supported usage

```rust
fn model_owner<C: gammalooprs::HasModel>(_context: &C) {}
```

`get_model`

Kind: `Method`

```
fn get_model(& self) -> & Model;
```

Exhaustive reference · Source

`Install interrupt handling`

`Installs GammaLoop's process interrupt handler before a long-running integration.`

`Installs GammaLoop's process interrupt handler before a long-running integration.`

```
fn set_interrupt_handler()
```

Supported usage

```
gammalooprs::set_interrupt_handler();
```

Exhaustive reference · Source

`gammaloop-api`

GammaLoop's stable facade and command API. Use this curated boundary for applications while consulting the full Rustdoc for exhaustive details.

`State loading options`

`Selects the state folder, boot commands, settings overrides, logging, and read-only mode.`

`Selects the state folder, boot commands, settings overrides, logging, and read-only mode.`

```
#[derive(Debug, Clone, PartialEq, Default)] pub struct StateLoadOption
{
    pub clean_state : bool, pub boot_commands_path : Option < PathBuf > , pub
    state_folder : Option < PathBuf > , pub model_file : Option < PathBuf > ,
    pub trace_logs_filename : Option < String > , pub level : Option <
    LogLevel > , pub logfile_level : Option < LogLevel > , pub logging_prefix
    : Option < LogFormat > , pub read_only_state : bool, pub
    settings_global_path : Option < PathBuf > , pub
    settings_runtime_defaults_path : Option < PathBuf > ,
}
```

Reference feature matrix: `cli` (item is available without an additional gate)

Supported usage

```
let options = gammaloop_api::StateLoadOption::default();
```

`clean_state`
Kind: `Field`

`bool`

`boot_commands_path`
Kind: `Field`

`Option < PathBuf >`

`state_folder`
Kind: `Field`

`Option < PathBuf >`

`model_file`
Kind: `Field`

`Option < PathBuf >`

`trace_logs_filename`
Kind: `Field`

`Option < String >`

`level`
Kind: `Field`

`Option < LogLevel >`

`logfile_level`
Kind: `Field`

`Option < LogLevel >`

`logging_prefix`
Kind: `Field`

`Option < LogFormat >`

`read_only_state`
Kind: `Field`

`bool`

settings_global_path
Kind: Field

`Option < PathBuf >`

settings_runtime_defaults_path
Kind: Field

`Option < PathBuf >`

Exhaustive reference · Source

Load a GammaLoop state
Loads the selected state and returns the supported session boundary.

Loads the selected state and returns the supported session boundary.

`fn load(self) -> Result < LoadedState >`

Reference feature matrix: `cli` (item is available without an additional gate)

Parameter	Type
self	self

Returns: `Result < LoadedState >`

Supported usage

```
let loaded = gammaloop_api::StateLoadOption::default().load()?;
```

Exhaustive reference · Source

Create a CLI session
Borrows a loaded state as a command session while keeping state ownership explicit.

Borrows a loaded state as a command session while keeping state ownership explicit.

`fn cli_session(& mut self) -> CliSession < '_ >`

Reference feature matrix: `cli` (item is available without an additional gate)

Parameter	Type
self	& mut self

Returns: `CliSession < '_ >`

Supported usage

```
let mut loaded = gammaloop_api::StateLoadOption::default().load()?; let session =
loaded.cli_session();
```

Exhaustive reference · Source

Command-line settings
Owns the generated command-line, global, session, and state-loading settings.

Owns the generated command-line, global, session, and state-loading settings.

```
#[derive(Debug, Clone, Deserialize, Serialize, JsonSchema)]
#[serde(default, deny_unknown_fields)] pub struct CLISettings
{
    #[serde(skip_serializing_if = "is_true")] pub try_strings : bool,
    #[serde(skip_serializing_if = "is_false")] pub override_state : bool,
```

```
    #[serde(skip_serializing_if = "IsDefault::is_default")] pub state :
    StateSettings, #[serde(skip_serializing_if = "IsDefault::is_default")] pub
    global : GlobalSettings, #[serde(skip)] #[schemars(skip)] pub session :
    SessionSettings,
}
```

Reference feature matrix: `cli` (item is available without an additional gate)

Supported usage

```
let settings = gammaloop_api::CLISettings::default();
```

`try_strings`
Kind: `Field`

`bool`

`override_state`
Kind: `Field`

`bool`

`state`
Kind: `Field`

`StateSettings`

`global`
Kind: `Field`

`GlobalSettings`

`session`
Kind: `Field`

`SessionSettings`

Exhaustive reference · Source

`GammaLoop command`

`Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.`

`Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.`

```
gammaloop [OPTIONS] [COMMAND]
```

Reference feature matrix: `cli` (item is available without an additional gate)

Supported usage

```
gammaloop --help
```

`try_strings`
Kind: `Field`

`bool`

`override_state`
Kind: `Field`

`bool`

state
Kind: Field

StateSettings

global
Kind: Field

GlobalSettings

session
Kind: Field

SessionSettings

Exhaustive reference · Source

gammaloop-python

Exports registered by gammaloop.

GammaLoopAPI

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

class GammaLoopAPI:

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```
from gammaloop._gammaloop import GammaLoopAPI
help(GammaLoopAPI)
```

__new__
Kind: AssociatedFunction

```
def __new__(cls, state_folder: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, boot_commands_path: typing.Optional[builtins.str | os.PathLike
| pathlib.Path] = None, model_file: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, trace_logs_filename: typing.Optional[builtins.str] = None,
level: typing.Optional[LogLevel] = None, logfile_level: typing.Optional[LogLevel] =
None, logging_prefix: builtins.object | None = None, read_only_state: builtins.bool
= ..., settings_global_path: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, settings_runtime_defaults_path: typing.Optional[builtins.str |
os.PathLike | pathlib.Path] = None, clean_state: builtins.bool = ...) ->
GammaLoopAPI:
```

state_folder
Kind: Parameter

state_folder: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

Default: None

boot_commands_path
Kind: Parameter

boot_commands_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

Default: None

`model_file`
Kind: `Parameter`

`model_file: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`

Default: `None`

`trace_logs_filename`
Kind: `Parameter`

`trace_logs_filename: typing.Optional[builtins.str] = None`

Default: `None`

`level`
Kind: `Parameter`

`level: typing.Optional[LogLevel] = None`

Default: `None`

`logfile_level`
Kind: `Parameter`

`logfile_level: typing.Optional[LogLevel] = None`

Default: `None`

`logging_prefix`
Kind: `Parameter`

`logging_prefix: builtins.object | None = None`

Default: `None`

`read_only_state`
Kind: `Parameter`

`read_only_state: builtins.bool = ...`

Default: `...`

`settings_global_path`
Kind: `Parameter`

`settings_global_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`

Default: `None`

`settings_runtime_defaults_path`
Kind: `Parameter`

`settings_runtime_defaults_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`

Default: `None`

`clean_state`
Kind: `Parameter`

`clean_state: builtins.bool = ...`

Default: ...

evaluate_sample

Kind: Method

```
def evaluate_sample(self, point: typing.Sequence[builtins.float], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weight: typing.Optional[builtins.float] = None, discrete_dim:
typing.Optional[typing.Sequence[builtins.int]] = None, graph_name:
typing.Optional[builtins.str] = None, orientation: typing.Optional[builtins.int] =
None) -> EvaluationResult:
```

point

Kind: Parameter

```
point: typing.Sequence[builtins.float]
```

process_id

Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

integrand_name

Kind: Parameter

```
integrand_name: typing.Optional[builtins.str] = None
```

Default: None

use_arb_prec

Kind: Parameter

```
use_arb_prec: builtins.bool = ...
```

Default: ...

minimal_output

Kind: Parameter

```
minimal_output: builtins.bool = ...
```

Default: ...

return_events

Kind: Parameter

```
return_events: typing.Optional[builtins.bool] = None
```

Default: None

momentum_space

Kind: Parameter

```
momentum_space: builtins.bool = ...
```

Default: ...

integrator_weight
Kind: Parameter

```
integrator_weight: typing.Optional[builtins.float] = None
```

Default: None

discrete_dim
Kind: Parameter

```
discrete_dim: typing.Optional[typing.Sequence[builtins.int]] = None
```

Default: None

graph_name
Kind: Parameter

```
graph_name: typing.Optional[builtins.str] = None
```

Default: None

orientation
Kind: Parameter

```
orientation: typing.Optional[builtins.int] = None
```

Default: None

evaluate_samples
Kind: Method

```
def evaluate_samples(self, points: numpy.typing.NDArray[numpy.float64], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] =
None, discrete_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None,
graph_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None,
orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None)
-> BatchEvaluationResult:
```

points
Kind: Parameter

```
points: numpy.typing.NDArray[numpy.float64]
```

process_id
Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

integrand_name
Kind: Parameter

```
integrand_name: typing.Optional[builtins.str] = None
```

Default: None

`use_arb_prec`
Kind: `Parameter`

```
use_arb_prec: builtins.bool = ...
```

Default: `...`

`minimal_output`
Kind: `Parameter`

```
minimal_output: builtins.bool = ...
```

Default: `...`

`return_events`
Kind: `Parameter`

```
return_events: typing.Optional[builtins.bool] = None
```

Default: `None`

`momentum_space`
Kind: `Parameter`

```
momentum_space: builtins.bool = ...
```

Default: `...`

`integrator_weights`
Kind: `Parameter`

```
integrator_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] = None
```

Default: `None`

`discrete_dims`
Kind: `Parameter`

```
discrete_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None
```

Default: `None`

`graph_names`
Kind: `Parameter`

```
graph_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None
```

Default: `None`

`orientations`
Kind: `Parameter`

```
orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None
```

Default: `None`

`import_graphs`
Kind: `Method`

```
def import_graphs(self, graphs: builtins.str, process_name:
typing.Optional[builtins.str] = None, process_id: typing.Optional[builtins.int] =
```

```
None, integrand_name: typing.Optional[builtins.str] = None, format: builtins.str
= ..., overwrite: builtins.bool = ..., append: builtins.bool = ...) -> None:
```

graphs

Kind: Parameter

```
graphs: builtins.str
```

process_name

Kind: Parameter

```
process_name: typing.Optional[builtins.str] = None
```

Default: None

process_id

Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

integrand_name

Kind: Parameter

```
integrand_name: typing.Optional[builtins.str] = None
```

Default: None

format

Kind: Parameter

```
format: builtins.str = ...
```

Default: ...

overwrite

Kind: Parameter

```
overwrite: builtins.bool = ...
```

Default: ...

append

Kind: Parameter

```
append: builtins.bool = ...
```

Default: ...

get_lmbs

Kind: Method

```
def get_lmbs(self, graphs: builtins.str, format: builtins.str = ...) ->
builtins.list[builtins.list[tuple[builtins.list[builtins.int],
builtins.list[builtins.int], builtins.dict[builtins.int,
tuple[builtins.list[builtins.int], builtins.list[builtins.int]]]]]]:
```

graphs

Kind: Parameter

```
graphs: builtins.str
```

format

Kind: Parameter

```
format: builtins.str = ...
```

Default: ...

get_orientations

Kind: Method

```
def get_orientations(self, graph_name: builtins.str, process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None) -> builtins.list[builtins.dict[builtins.int, builtins.int]]:
```

graph_name

Kind: Parameter

```
graph_name: builtins.str
```

process_id

Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

integrand_name

Kind: Parameter

```
integrand_name: typing.Optional[builtins.str] = None
```

Default: None

get_model

Kind: Method

```
def get_model(self) -> builtins.str:
```

evaluate

Kind: Method

```
def evaluate(self, process_id: typing.Optional[builtins.int] = None,
graphs_group_name: typing.Optional[builtins.str] = None, result_path:
typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None, numerical:
builtins.bool = ..., number_of_terms_in_epsilon_expansion:
typing.Optional[builtins.int] = None) -> builtins.str:
```

process_id

Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

graphs_group_name

Kind: Parameter

```
graphs_group_name: typing.Optional[builtins.str] = None
```

Default: None

result_path
Kind: Parameter

```
result_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None
```

Default: None

numerical
Kind: Parameter

```
numerical: builtins.bool = ...
```

Default: ...

number_of_terms_in_epsilon_expansion
Kind: Parameter

```
number_of_terms_in_epsilon_expansion: typing.Optional[builtins.int] = None
```

Default: None

import_model
Kind: Method

```
def import_model(self, model_specifier: builtins.str | os.PathLike | pathlib.Path,
simplify_model: builtins.bool = ...) -> None:
```

model_specifier
Kind: Parameter

```
model_specifier: builtins.str | os.PathLike | pathlib.Path
```

simplify_model
Kind: Parameter

```
simplify_model: builtins.bool = ...
```

Default: ...

list_outputs
Kind: Method

```
def list_outputs(self) -> tuple[builtins.dict[builtins.str, builtins.int],
builtins.dict[builtins.str, builtins.int]]:
```

get_integrand_info
Kind: Method

```
def get_integrand_info(self, process_id: typing.Optional[builtins.int] = None,
integrand_name: typing.Optional[builtins.str] = None) -> IntegrandInfo:
```

process_id
Kind: Parameter

```
process_id: typing.Optional[builtins.int] = None
```

Default: None

integrand_name
Kind: Parameter

`integrand_name: typing.Optional[builtins.str] = None`

Default: `None`

`get_integrand_settings`
Kind: `Method`

```
def get_integrand_settings(self, process_id: typing.Optional[builtins.int] = None,
integrand_name: typing.Optional[builtins.str] = None) -> SettingsValue:
```

`process_id`
Kind: `Parameter`

`process_id: typing.Optional[builtins.int] = None`

Default: `None`

`integrand_name`
Kind: `Parameter`

`integrand_name: typing.Optional[builtins.str] = None`

Default: `None`

`get_run_history`
Kind: `Method`

```
def get_run_history(self) -> builtins.str:
```

`get_global_settings`
Kind: `Method`

```
def get_global_settings(self) -> builtins.str:
```

`get_active_command_blocks`
Kind: `Method`

```
def get_active_command_blocks(self) -> builtins.dict[builtins.str,
builtins.list[builtins.str]]:
```

`get_default_runtime_settings`
Kind: `Method`

```
def get_default_runtime_settings(self) -> SettingsValue:
```

`get_dot_files`
Kind: `Method`

```
def get_dot_files(self, process: typing.Optional[builtins.int | builtins.str] = None,
integrand_name: typing.Optional[builtins.str] = None, settings: DotExportSettings
= ...) -> builtins.str:
```

`process`
Kind: `Parameter`

`process: typing.Optional[builtins.int | builtins.str] = None`

Default: `None`

`integrand_name`
Kind: `Parameter`

```
integrand_name: typing.Optional[builtins.str] = None
```

Default: `None`

`settings`
Kind: `Parameter`

```
settings: DotExportSettings = ...
```

Default: `...`

`run`
Kind: `Method`

```
def run(self, command: builtins.str) -> None:
```

`command`
Kind: `Parameter`

```
command: builtins.str
```

`generate_cff`
Kind: `Method`

```
def generate_cff(self, dot_string: builtins.str, subgraph_nodes:
typing.Sequence[builtins.str], reverse_dangling: typing.Sequence[builtins.int],
orientation_pattern: typing.Optional[builtins.str] = None) ->
builtins.list[tuple[builtins.dict[builtins.int, builtins.int], builtins.str]]:
```

`dot_string`
Kind: `Parameter`

```
dot_string: builtins.str
```

`subgraph_nodes`
Kind: `Parameter`

```
subgraph_nodes: typing.Sequence[builtins.str]
```

`reverse_dangling`
Kind: `Parameter`

```
reverse_dangling: typing.Sequence[builtins.int]
```

`orientation_pattern`
Kind: `Parameter`

```
orientation_pattern: typing.Optional[builtins.str] = None
```

Default: `None`

`generate_cff_as_json_string`
Kind: `Method`

```
def generate_cff_as_json_string(self, dot_string: builtins.str, subgraph_nodes:
typing.Sequence[builtins.str], reverse_dangling: typing.Sequence[builtins.int],
orientation_pattern: typing.Optional[builtins.str] = None) -> builtins.str:
```

`dot_string`
Kind: `Parameter`

```
dot_string: builtins.str
```

subgraph_nodes
Kind: Parameter

```
subgraph_nodes: typing.Sequence[builtins.str]
```

reverse_dangling
Kind: Parameter

```
reverse_dangling: typing.Sequence[builtins.int]
```

orientation_pattern
Kind: Parameter

```
orientation_pattern: typing.Optional[builtins.str] = None
```

Default: None

Exhaustive reference · Source

atom_to_canonical_string
Parse a Symbolica expression and return its canonical string form.

Parse a Symbolica expression and return its canonical string form.

```
def atom_to_canonical_string(atom_str: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```python
from gammaloop._gammaloop import atom_to_canonical_string
help(atom_to_canonical_string)
```

Exhaustive reference · Source

evaluate_graph_overall_factor
Evaluate a graph's symbolic overall factor and return its canonical form.

Evaluate a graph's symbolic overall factor and return its canonical form.

```
def evaluate_graph_overall_factor(overall_factor: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```python
from gammaloop._gammaloop import evaluate_graph_overall_factor
help(evaluate_graph_overall_factor)
```

Exhaustive reference · Source

to_dots
Rewrite a Symbolica tensor expression into Idenso dot-product notation.

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

```
def to_dots(atom_str: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```python
from gammaloop._gammaloop import to_dots
help(to_dots)
```

Generated CLI and settings reference

The following tables come from the compiled Clap command factory, Schemars, and real serialized defaults.

Command tree

Command	Description
`gammaLoop`	`An implementation of the Local Unitarity method for computing differential collider cross-sections.`
`gammaLoop display`	
`gammaLoop display model`	
`gammaLoop display processes`	
`gammaLoop display integrand`	
`gammaLoop display quantities`	
`gammaLoop display observables`	
`gammaLoop display selectors`	
`gammaLoop display command_block`	
`gammaLoop display settings`	
`gammaLoop display settings global`	
`gammaLoop display settings default-runtime`	
`gammaLoop display settings process`	
`gammaLoop display settings help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop display settings help global`	
`gammaLoop display settings help default-runtime`	
`gammaLoop display settings help process`	
`gammaLoop display settings help help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop display help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop display help model`	
`gammaLoop display help processes`	
`gammaLoop display help integrand`	
`gammaLoop display help quantities`	
`gammaLoop display help observables`	
`gammaLoop display help selectors`	
`gammaLoop display help command_block`	

Command	Description
gammaLoop display help settings	
gammaLoop display help settings global	
gammaLoop display help settings default-runtime	
gammaLoop display help settings process	
gammaLoop display help help	Print this message or the help of the given subcommand(s)
gammaLoop duplicate	
gammaLoop duplicate integrand	
gammaLoop duplicate help	Print this message or the help of the given subcommand(s)
gammaLoop duplicate help integrand	
gammaLoop duplicate help help	Print this message or the help of the given subcommand(s)
gammaLoop set	
gammaLoop set base-dir	Set the base directory for gammaloop state and logs
gammaLoop set global	Set GLOBAL settings
gammaLoop set global file	Load from a settings file
gammaLoop set global string	Load settings from TOML content passed as a CLI string
gammaLoop set global kv	Set one or more dotted key-paths
gammaLoop set global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set global stored	Use the stored settings file
gammaLoop set global help	Print this message or the help of the given subcommand(s)
gammaLoop set global help file	Load from a settings file
gammaLoop set global help string	Load settings from TOML content passed as a CLI string
gammaLoop set global help kv	Set one or more dotted key-paths
gammaLoop set global help defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set global help stored	Use the stored settings file
gammaLoop set global help help	Print this message or the help of the given subcommand(s)
gammaLoop set default-runtime	Set DEFAULT RUNTIME settings
gammaLoop set default-runtime file	Load from a settings file
gammaLoop set default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop set default-runtime kv	Set one or more dotted key-paths

Command	**Description**
`gammaLoop set default-runtime defaults`	`Sync process runtime settings from the current default runtime settings`
`gammaLoop set default-runtime stored`	`Use the stored settings file`
`gammaLoop set default-runtime help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop set default-runtime help file`	`Load from a settings file`
`gammaLoop set default-runtime help string`	`Load settings from TOML content passed as a CLI string`
`gammaLoop set default-runtime help kv`	`Set one or more dotted key-paths`
`gammaLoop set default-runtime help defaults`	`Sync process runtime settings from the current default runtime settings`
`gammaLoop set default-runtime help stored`	`Use the stored settings file`
`gammaLoop set default-runtime help help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop set model`	`Set Model parameters`
`gammaLoop set process`	`Set settings for a PROCESS`
`gammaLoop set process file`	`Load from a settings file`
`gammaLoop set process string`	`Load settings from TOML content passed as a CLI string`
`gammaLoop set process kv`	`Set one or more dotted key-paths`
`gammaLoop set process defaults`	`Sync process runtime settings from the current default runtime settings`
`gammaLoop set process stored`	`Use the stored settings file`
`gammaLoop set process add`	`Add a named quantity/observable/selector`
`gammaLoop set process add quantity`	
`gammaLoop set process add observable`	
`gammaLoop set process add selector`	
`gammaLoop set process add help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop set process add help quantity`	
`gammaLoop set process add help observable`	
`gammaLoop set process add help selector`	
`gammaLoop set process add help help`	`Print this message or the help of the given subcommand(s)`
`gammaLoop set process update`	`Update a named quantity/observable/selector`

Command	Description
gammaLoop set process update quantity	
gammaLoop set process update observable	
gammaLoop set process update selector	
gammaLoop set process update help	Print this message or the help of the given subcommand(s)
gammaLoop set process update help quantity	
gammaLoop set process update help observable	
gammaLoop set process update help selector	
gammaLoop set process update help help	Print this message or the help of the given subcommand(s)
gammaLoop set process remove	Remove a named quantity/observable/selector
gammaLoop set process remove quantity	
gammaLoop set process remove observable	
gammaLoop set process remove selector	
gammaLoop set process remove help	Print this message or the help of the given subcommand(s)
gammaLoop set process remove help quantity	
gammaLoop set process remove help observable	
gammaLoop set process remove help selector	
gammaLoop set process remove help help	Print this message or the help of the given subcommand(s)
gammaLoop set process help	Print this message or the help of the given subcommand(s)
gammaLoop set process help file	Load from a settings file
gammaLoop set process help string	Load settings from TOML content passed as a CLI string
gammaLoop set process help kv	Set one or more dotted key-paths
gammaLoop set process help defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set process help stored	Use the stored settings file
gammaLoop set process help add	Add a named quantity/observable/selector
gammaLoop set process help add quantity	

Command	Description
gammaLoop set process help add observable	
gammaLoop set process help add selector	
gammaLoop set process help update	Update a named quantity/observable/selector
gammaLoop set process help update quantity	
gammaLoop set process help update observable	
gammaLoop set process help update selector	
gammaLoop set process help remove	Remove a named quantity/observable/selector
gammaLoop set process help remove quantity	
gammaLoop set process help remove observable	
gammaLoop set process help remove selector	
gammaLoop set process help help	Print this message or the help of the given subcommand(s)
gammaLoop set help	Print this message or the help of the given subcommand(s)
gammaLoop set help base-dir	Set the base directory for gammaloop state and logs
gammaLoop set help global	Set GLOBAL settings
gammaLoop set help global file	Load from a settings file
gammaLoop set help global string	Load settings from TOML content passed as a CLI string
gammaLoop set help global kv	Set one or more dotted key-paths
gammaLoop set help global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help global stored	Use the stored settings file
gammaLoop set help default-runtime	Set DEFAULT RUNTIME settings
gammaLoop set help default-runtime file	Load from a settings file
gammaLoop set help default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop set help default-runtime kv	Set one or more dotted key-paths
gammaLoop set help default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help default-runtime stored	Use the stored settings file

Command	Description
gammaLoop set help model	Set Model parameters
gammaLoop set help process	Set settings for a PROCESS
gammaLoop set help process file	Load from a settings file
gammaLoop set help process string	Load settings from TOML content passed as a CLI string
gammaLoop set help process kv	Set one or more dotted key-paths
gammaLoop set help process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help process stored	Use the stored settings file
gammaLoop set help process add	Add a named quantity/observable/selector
gammaLoop set help process add quantity	
gammaLoop set help process add observable	
gammaLoop set help process add selector	
gammaLoop set help process update	Update a named quantity/observable/selector
gammaLoop set help process update quantity	
gammaLoop set help process update observable	
gammaLoop set help process update selector	
gammaLoop set help process remove	Remove a named quantity/observable/selector
gammaLoop set help process remove quantity	
gammaLoop set help process remove observable	
gammaLoop set help process remove selector	
gammaLoop set help help	Print this message or the help of the given subcommand(s)
gammaLoop import	
gammaLoop import model	Generate integrands
gammaLoop import graphs	
gammaLoop import help	Print this message or the help of the given subcommand(s)
gammaLoop import help model	Generate integrands
gammaLoop import help graphs	
gammaLoop import help help	Print this message or the help of the given subcommand(s)
gammaLoop save	

Command	Description
gammaLoop save dot	
gammaLoop save uv-forest	
gammaLoop save standalone	
gammaLoop save state	
gammaLoop save schema	regenerate the schema files
gammaLoop save help	Print this message or the help of the given subcommand(s)
gammaLoop save help dot	
gammaLoop save help uv-forest	
gammaLoop save help standalone	
gammaLoop save help state	
gammaLoop save help schema	regenerate the schema files
gammaLoop save help help	Print this message or the help of the given subcommand(s)
gammaLoop run	
gammaLoop start_commands_block	
gammaLoop finish_commands_block	
gammaLoop remove	
gammaLoop remove processes	
gammaLoop remove help	Print this message or the help of the given subcommand(s)
gammaLoop remove help processes	
gammaLoop remove help help	Print this message or the help of the given subcommand(s)
gammaLoop integrate	
gammaLoop generate	Generate integrands
gammaLoop generate xs	Cross-section (forward-scattering) generation
gammaLoop generate amp	Amplitude generation
gammaLoop generate existing	Reuse an already generated process
gammaLoop generate help	Print this message or the help of the given subcommand(s)
gammaLoop generate help xs	Cross-section (forward-scattering) generation
gammaLoop generate help amp	Amplitude generation
gammaLoop generate help existing	Reuse an already generated process
gammaLoop generate help help	Print this message or the help of the given subcommand(s)
gammaLoop select	
gammaLoop quit	Quit gammaloop
gammaLoop inspect	Inspect a single phase-space point / momentum configuration
gammaLoop approach	Approach a phase-space point along one or more axes

Command	Description
gammaLoop evaluate	
gammaLoop renormalize	
gammaLoop bench	Benchmark raw integrand evaluation speed
gammaLoop profile	
gammaLoop profile ultra-violet	Ultraviolet profile analysis
gammaLoop profile bulk	Bulk profile analysis
gammaLoop profile help	Print this message or the help of the given subcommand(s)
gammaLoop profile help ultra-violet	Ultraviolet profile analysis
gammaLoop profile help bulk	Bulk profile analysis
gammaLoop profile help help	Print this message or the help of the given subcommand(s)
gammaLoop batch	HPC batch evaluation branch
gammaLoop !	
gammaLoop help	Print this message or the help of the given subcommand(s)
gammaLoop help display	
gammaLoop help display model	
gammaLoop help display processes	
gammaLoop help display integrand	
gammaLoop help display quantities	
gammaLoop help display observables	
gammaLoop help display selectors	
gammaLoop help display command_block	
gammaLoop help display settings	
gammaLoop help display settings global	
gammaLoop help display settings default-runtime	
gammaLoop help display settings process	
gammaLoop help duplicate	
gammaLoop help duplicate integrand	
gammaLoop help set	
gammaLoop help set base-dir	Set the base directory for gammaloop state and logs
gammaLoop help set global	Set GLOBAL settings
gammaLoop help set global file	Load from a settings file

Command	Description
gammaLoop help set global string	Load settings from TOML content passed as a CLI string
gammaLoop help set global kv	Set one or more dotted key-paths
gammaLoop help set global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set global stored	Use the stored settings file
gammaLoop help set default-runtime	Set DEFAULT RUNTIME settings
gammaLoop help set default-runtime file	Load from a settings file
gammaLoop help set default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop help set default-runtime kv	Set one or more dotted key-paths
gammaLoop help set default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set default-runtime stored	Use the stored settings file
gammaLoop help set model	Set Model parameters
gammaLoop help set process	Set settings for a PROCESS
gammaLoop help set process file	Load from a settings file
gammaLoop help set process string	Load settings from TOML content passed as a CLI string
gammaLoop help set process kv	Set one or more dotted key-paths
gammaLoop help set process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set process stored	Use the stored settings file
gammaLoop help set process add	Add a named quantity/observable/selector
gammaLoop help set process add quantity	
gammaLoop help set process add observable	
gammaLoop help set process add selector	
gammaLoop help set process update	Update a named quantity/observable/selector
gammaLoop help set process update quantity	
gammaLoop help set process update observable	
gammaLoop help set process update selector	
gammaLoop help set process remove	Remove a named quantity/observable/selector

Command	Description
gammaLoop help set process remove quantity	
gammaLoop help set process remove observable	
gammaLoop help set process remove selector	
gammaLoop help import	
gammaLoop help import model	Generate integrands
gammaLoop help import graphs	
gammaLoop help save	
gammaLoop help save dot	
gammaLoop help save uv-forest	
gammaLoop help save standalone	
gammaLoop help save state	
gammaLoop help save schema	regenerate the schema files
gammaLoop help run	
gammaLoop help start_commands_block	
gammaLoop help finish_commands_block	
gammaLoop help remove	
gammaLoop help remove processes	
gammaLoop help integrate	
gammaLoop help generate	Generate integrands
gammaLoop help generate xs	Cross-section (forward-scattering) generation
gammaLoop help generate amp	Amplitude generation
gammaLoop help generate existing	Reuse an already generated process
gammaLoop help select	
gammaLoop help quit	Quit gammaloop
gammaLoop help inspect	Inspect a single phase-space point / momentum configuration
gammaLoop help approach	Approach a phase-space point along one or more axes
gammaLoop help evaluate	
gammaLoop help renormalize	
gammaLoop help bench	Benchmark raw integrand evaluation speed
gammaLoop help profile	
gammaLoop help profile ultra-violet	Ultraviolet profile analysis
gammaLoop help profile bulk	Bulk profile analysis
gammaLoop help batch	HPC batch evaluation branch
gammaLoop help !	

Command	Description
gammaLoop help help	Print this message or the help of the given subcommand(s)

Settings

Path	Type	Default
cli.global.display_directive	string	"info"
cli.global.generation.compile.compilation_mode	string	"symjit"
cli.global.generation.compile.compiler	string	"g++"
cli.global.generation.compile.custom	array	[]
cli.global.generation.compile.fast_math	boolean	true
cli.global.generation.compile.optimization_level	string	"O2"
cli.global.generation.compile.unsafe_math	boolean	true
cli.global.generation.evaluator.abort_level	integer	0
cli.global.generation.evaluator.compile	boolean	false
cli.global.generation.evaluator.cpe_iterations	integer | null	no serialized default
cli.global.generation.evaluator.direct_translation	boolean	true
cli.global.generation.evaluator.do_algebra	boolean	false
cli.global.generation.evaluator.do_fn_map_replacements	boolean	false
cli.global.generation.evaluator.horner_iterations	integer	1
cli.global.generation.evaluator.iterative_orientation_optimization	boolean	true
cli.global.generation.evaluator.max_common_pair_cache_entries	integer	1000000
cli.global.generation.evaluator.max_common_pair_distance	integer	1000
cli.global.generation.evaluator.max_horner_scheme_variables	integer	500
cli.global.generation.evaluator.n_cores	integer	1
cli.global.generation.evaluator.sparse_execution_mode	array	no serialized default
cli.global.generation.evaluator.store_atom	boolean	false
cli.global.generation.evaluator.summed	boolean	false
cli.global.generation.evaluator.summed_function_map	boolean	false
cli.global.generation.evaluator.tensor_network_contraction_order	string	"sparse_order_aware"
cli.global.generation.evaluator.verbose	boolean	false
cli.global.generation.feyngen.gamma_simplification_closure_check	boolean	false
cli.global.generation.force_cuts	array	[]
cli.global.generation.orientation_pattern	null | string	no serialized default
cli.global.generation.override_lmb_heuristics	boolean	false
cli.global.generation.threshold_subtraction.assume_positive_external_energies	boolean	true
cli.global.generation.threshold_subtraction.check_esurface_at_generation	boolean	false
cli.global.generation.threshold_subtraction.disable_integrated_ct	boolean	false
cli.global.generation.threshold_subtraction.enable_thresholds	boolean	true
cli.global.generation.threshold_subtraction.esurface_existence_threshold	number	1e-7
cli.global.generation.threshold_subtraction.skip_thresholds_that_are_cuts	boolean	true
cli.global.generation.tropical_subgraph_table.disable_tropical_generation	boolean	false

Path	Type	Default
cli.global.generation.tropical_subgraph_table.panic_on_fail	boolean	false
cli.global.generation.tropical_subgraph_table.target_omega	number	1.0
cli.global.generation.uv.add_marker	boolean	false
cli.global.generation.uv.final_integrand	string	"ThreeD"
cli.global.generation.uv.generate_integrated	boolean	true
cli.global.generation.uv.inner_products	boolean	true
cli.global.generation.uv.keep_marker	boolean	true
cli.global.generation.uv.orchestrator	string	"hedge_poset"
cli.global.generation.uv.renormalization_prescription.log_divergent	string	"MUV"
cli.global.generation.uv.renormalization_prescription.massive_power_divergent	string	"MUV"
cli.global.generation.uv.renormalization_prescription.massless_power_divergent	string	"MUV"
cli.global.generation.uv.renormalization_prescription.overrides	array	[]
cli.global.generation.uv.softct	boolean	true
cli.global.generation.uv.subtract_uv	boolean	true
cli.global.generation.uv.vakint.additional_normalization	string	"1"
cli.global.generation.uv.vakint.alphaloop.susbstitute_masters	boolean	true
cli.global.generation.uv.vakint.clean_tmp_dir	boolean	true
cli.global.generation.uv.vakint.evaluation_methods	array	["alphaloop","matad","fmft"]
cli.global.generation.uv.vakint.fmft.expand_masters	boolean	true
cli.global.generation.uv.vakint.fmft.susbstitute_masters	boolean	true
cli.global.generation.uv.vakint.form_exe_path	string	"form"
cli.global.generation.uv.vakint.matad.direct_numerical_substition	boolean	true
cli.global.generation.uv.vakint.matad.expand_masters	boolean	true
cli.global.generation.uv.vakint.matad.substitute_hpls	boolean	true
cli.global.generation.uv.vakint.matad.susbstitute_masters	boolean	true
cli.global.generation.uv.vakint.normalization	string	"MSbar"
cli.global.generation.uv.vakint.pysecdec.max_n_evals	integer	1000000000000
cli.global.generation.uv.vakint.pysecdec.min_n_evals	integer	10000
cli.global.generation.uv.vakint.pysecdec.quiet	boolean	true
cli.global.generation.uv.vakint.pysecdec.relative_precision	number	1e-7
cli.global.generation.uv.vakint.pysecdec.reuse_existing_output	string | null	no serialized default
cli.global.generation.uv.vakint.python_exe_path	string	"python3"
cli.global.generation.uv.vakint.run_time_decimal_precision	integer	100
cli.global.generation.uv.vakint.temporary_directory	string | null	no serialized default
cli.global.generation.vector_polarization_sum_gauge	string	"LightLikeAxial"
cli.global.log_style.full_line_source	boolean	false
cli.global.log_style.include_fields	boolean	false
cli.global.log_style.log_format	string	"Long"
cli.global.log_style.short_timestamp	boolean	false
cli.global.logfile_directive	string	"off"

Path	Type	Default
cli.global.n_cores.compile	integer	1
cli.global.n_cores.feyngen	integer	1
cli.global.n_cores.generate	integer	1
cli.global.n_cores.integrate	integer	1
cli.override_state	boolean	false
cli.state.folder	string	"./gammaloop_state"
cli.state.name	string | null	""
cli.try_strings	boolean	true
runtime.general.additional_param_values	array	[]
runtime.general.debug_cache	boolean	false
runtime.general.disable_flux_factor	boolean	false
runtime.general.enable_cache	boolean	false
runtime.general.evaluator_method	string	"SingleParametric"
runtime.general.generate_events	boolean	false
runtime.general.integral_unit	string	"auto"
runtime.general.load_compiled_cff	boolean	false
runtime.general.m_uv	number	1000.0
runtime.general.mu_r	number	1000.0
runtime.general.orientation_pat.pat	null | string	no serialized default
runtime.general.renormalization_localization_scale	number	1000.0
runtime.general.store_additional_weights_in_event	boolean	false
runtime.general.use_ltd	boolean	false
runtime.h_function.enabled_dampening	boolean	true
runtime.h_function.function	string	"poly_exponential"
runtime.h_function.power	integer | null	no serialized default
runtime.h_function.sigma	number	1.0
runtime.integrand.n_3d_momenta	integer	no serialized default
runtime.integrator.bin_number_evolution	array | null	no serialized default
runtime.integrator.continuous_dim_learning_rate	number	1.0
runtime.integrator.discrete_dim_learning_rate	number	1.0
runtime.integrator.integrated_phase	string	"real"
runtime.integrator.max_prob_ratio	number	30.0
runtime.integrator.min_samples_for_update	integer	1000
runtime.integrator.n_bins	integer	64
runtime.integrator.n_increase	integer	10000
runtime.integrator.n_max	integer	10000000000
runtime.integrator.n_start	integer	100000
runtime.integrator.observables_output_format	array	["json"]
runtime.integrator.seed	integer	69
runtime.integrator.show_max_wgt_info	boolean	true

Path	Type	Default
runtime.integrator.target_absolute_accuracy	number | null	no serialized default
runtime.integrator.target_relative_accuracy	number | null	no serialized default
runtime.integrator.train_on_avg	boolean	false
runtime.kinematics.e_cm	number	64.0
runtime.kinematics.externals.data.helicities	array	[]
runtime.kinematics.externals.data.improvement_settings.large_deformation_check	null | number	1e-12
runtime.kinematics.externals.data.improvement_settings.mode	string	"none"
runtime.kinematics.externals.data.improvement_settings.only_warn_on_large_deformation	boolean	false
runtime.kinematics.externals.data.momenta	array	[]
runtime.kinematics.externals.type	string	"constant"
runtime.model	object	{}
runtime.observables	object	{}
runtime.quantities	object	{}
runtime.sampling.alpha	number	3.0
runtime.sampling.b	number	1.0
runtime.sampling.coordinate_system	string	"spherical"
runtime.sampling.graph_names	array	[]
runtime.sampling.graphs	string	"summed"
runtime.sampling.lmb_basis_ids	object	no serialized default
runtime.sampling.lmb_channel_weight	string	"ose"
runtime.sampling.lmb_channels	string	"summed"
runtime.sampling.lmb_multichanneling	boolean	false
runtime.sampling.mapping	string	"linear"
runtime.sampling.orientations	string	"summed"
runtime.sampling.power	number	1.0
runtime.selectors	object	{}
runtime.stability.check_on_norm	boolean	true
runtime.stability.escalate_if_exact_zero	boolean	false
runtime.stability.levels	array	[{"escalate_for_large_weight_threshold":0.9,"pr {"escalate_for_large_weight_threshold":-1.0,"pr {"escalate_for_large_weight_threshold":-1.0,"pr
runtime.stability.loop_momenta_norm_escalation_factor	number	-1.0
runtime.stability.recording.record_all_stability_levels	boolean	no serialized default
runtime.stability.recording.record_loop_momenta_escalation	boolean	no serialized default
runtime.stability.recording.record_rotated_results	boolean	no serialized default
runtime.stability.rotation_axis	array	[{"alpha":0.1,"beta":0.2,"gamma":0.3,"type":"eu
runtime.subtraction.disable_threshold_subtraction	boolean	false
runtime.subtraction.esurface_existence_threshold	number	1e-7
runtime.subtraction.integrated_ct_settings.range.h_function_settings.enabled_dampening	boolean	true
runtime.subtraction.integrated_ct_settings.range.h_function_settings.function	string	"poly_exponential"
runtime.subtraction.integrated_ct_settings.range.h_function_settings.power	integer | null	no serialized default

Path	Type	Default
runtime.subtraction.integrated_ct_settings.range.h_function_settings.sigma	number	1.0
runtime.subtraction.integrated_ct_settings.range.type	string	"infinite"
runtime.subtraction.local_ct_settings.uv_localisation.dynamic_width	boolean	false
runtime.subtraction.local_ct_settings.uv_localisation.force_uv_dampers_to_one	boolean	false
runtime.subtraction.local_ct_settings.uv_localisation.gaussian_width	number	1.0
runtime.subtraction.local_ct_settings.uv_localisation.sliver_width	number	10.0
runtime.subtraction.overlap_settings.check_global_center	boolean	true
runtime.subtraction.overlap_settings.force_global_center	array (nullable)	not serialized default
runtime.subtraction.overlap_settings.try_origin	boolean	true
runtime.subtraction.overlap_settings.try_origin_all_lmbs	boolean	false
runtime.subtraction.radial_root_residual_tolerance	number	64.0

Snapshot and version metadata

This manual was rendered from one immutable workspace identity.

- **Channel:** `latest`
- **Route:** `/products/gammaloop/latest/`
- **Git commit:** `c169d49f8084f7b7f4b23e647729c7030ba4eca2`

Workspace component versions

- `gammaloop/gammaloop-rs` — `0.3.4` (`features: none (Cargo defaults disabled)`)
- `gammaloop/gammaloop-api` — `0.1.0` (`features: cli (Cargo defaults disabled; complete explicit matrix)`)
- `gammaloop/gammaloop` — `0.3.4` (`features: python_api, python_abi, python_stubgen (Cargo defaults disabled; complete explicit matrix)`)
- `linnet/linnet` — `0.17.0` (`features: nodestore-vec, serde, bincode, drawing, symbolica (Cargo defaults disabled; complete explicit matrix)`)
- `linnet/linnet-py` — `0.1.0` (`features: extension-module, abi3-py310 (Cargo defaults disabled; complete explicit matrix)`)
- `spenso/spenso` — `0.6.0` (`features: shadowing (Cargo defaults disabled; complete explicit matrix)`)
- `spenso/spenso-macros` — `0.3.2` (`features: shadowing (Cargo defaults disabled; complete explicit matrix)`)
- `spenso/spenso-hep-lib` — `0.1.7` (`features: none (Cargo defaults disabled)`)
- `spenso/spynso3` — `0.1.2` (`features: python_stubgen (Cargo defaults disabled; complete explicit matrix)`)
- `idenso/idenso` — `0.3.0` (`features: bincode, reference-cases (Cargo defaults disabled; complete explicit matrix)`)
- `idenso/idenso` — `0.3.0` (`features: python, python_stubgen (Cargo defaults disabled; complete explicit matrix)`)
- `vakint/vakint` — `0.1.2` (`features: none (Cargo defaults disabled)`)
- `vakint/vakint` — `0.1.2` (`features: symbolica_community_module, python_stubgen (Cargo defaults disabled; complete explicit matrix)`)